INCH-POUND
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SUPERSEDING
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PERFORMANCE SPECIFICATION

PUMP, ROTARY, HAND-DRIVEN, AND MOTOR, HYDRAULIC-REMOTE VALVE OPERATION, NAVAL SHIPBOARD USE

Reinstated after 08 December 2020 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand-driven, hydraulic, rotary pump, and motor equipment intended for service limited to the remote operation of valves for naval shipboard use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-15024	-	Plates and Tags for Identification of Equipment, General Specification for
MIL-PRF-17672	-	Hydraulic Fluid, Petroleum, Inhibited

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2193	-	Ship Hydraulic System Components

(Copies of these documents are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SAE INTERNATIONAL

SAE J1926/1 - Connections for General Use and Fluid Power – Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing – Part 1: Threaded Port with O-Ring Seal in Truncated Housing

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample pump and motor unit shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Materials. Unless otherwise specified herein, materials shall be selected from those specified in MIL-STD-2193, compatible with hydraulic fluid (2135 T-H) as specified in MIL-PRF-17672, and for the purpose intended, under the conditions of temperature and shock specified herein. The choice of materials shall be such that the parts will not gall, corrode, or stick after periods of inactivity at the temperatures specified (see 3.9). Cast iron shall not be used.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Pump.

3.3.1 Design. The hydraulic pump shall be a rotary, positive displacement, crank operated type for clockwise rotation. The pump shall be provided with mounting lugs or brackets for bolting to the ship structure.

3.3.2 Performance. The hydraulic pump shall be capable of a discharge pressure of 350 and 900 pounds per square inch (psi) (24 and 62 bar), with a crank speed of 60 revolutions per minute (rpm) and crank input torques of 42 and 83 foot-pounds, respectively (57 and 112 newton-meters [N•m]).

3.4 Hydraulic motor.

3.4.1 Design. The motor shall consist of an assembly with inlet and outlet ports and having the drive shaft projecting for connection to a valve stem or gear unit for driving a valve stem. High-pressure hydraulic fluid shall enter through the motor inlet, causing the drive shaft of the motor to rotate, and shall pass out through the motor outlet at a low pressure. Motor reversal shall be accomplished by the use of a four-way selector cock that diverts the fluid to the desired port of the motor and shall be located in the hydraulic system near the pump.

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3.4.2 Performance. With inlet pressures at 350 psi and 900 psi (24 and 62 bar), the output torque of the motor drive shaft shall be not less than 30 foot-pounds and 60 foot-pounds, respectively (41 and 81 N•m), with the fluid temperature not less than 50 °F (10 °C).

3.5 Housings. Pump and motor housings shall be designed for a working pressure of 750 psi (52 bar). Test pressure shall be 1,000 psi (69 bar).

3.6 Hydraulic ports. Hydraulic connections for the pump and motor housings shall be straight thread SAE J1926/1 O-ring seal. Adapters to system piping shall be male union type with one end a straight thread O-ring boss and the other an O-ring face seal.

3.7 Pump crank. The crank shall be a design having an effective radius of 10 inches (254 millimeters [mm]), with an 8-inch (203 mm) handle for applying both hands. The crank shall be easily detached and shall incorporate a hub that latches to the pump drive shaft while in use.

3.8 Seals. The pump and motor shall be provided with seals around the shaft bearings and at all other joints or openings to prevent leakage of hydraulic fluid out of the unit when subjected to an internal hydraulic pressure of not less than 1,000 psi (69 bar).

3.9 Operating temperatures. The hydraulic units shall operate satisfactorily for all temperatures of the hydraulic fluid from 50 to 200 °F (10 to 93 °C).

3.10 Hydraulic fluid. The hydraulic fluid used in this equipment shall be in accordance with symbol 2135 T-H as specified in MIL-PRF-17672

3.11 Shock resistance. The pump and motor shall not become inoperative or sustain damage when subjected to the grade A, class I equipment shock test as specified in MIL-DTL-901 (see 4.5.6).

3.12 Marking. Each unit shall be permanently and legibly marked with an identification plate conforming to type A, B, C, D, or L as specified in MIL-DTL-15024, with the manufacturer's name or trademark, type or model number, and the national stock number.

3.13 Workmanship. The entire pump and motor unit shall be thoroughly cleaned both inside and out. Valves shall be free from burrs, loose sand, grit, scale, metal shavings, and all other contaminants that can affect performance and damage sealing surfaces. There shall be no sharp edges. All sealing, rubbing, and coated surfaces shall be smooth with no blemishes, scratches, scrapes, or gouges.

3.14 Special tools. Special tools shall not be required for servicing the pump and motor units. Special tools are defined as tools not listed in the GSA Global Supply Catalog.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the first pump and motor unit of each new design when a first article sample is required (see 3.1). This inspection shall include the examinations and tests specified in table I. Some tests may be able to be conducted concurrently; however, the shock test shall be performed separately.

4.3 Conformance inspection. All supplied units shall have the conformance tests specified in table I, or as specified (see 6.2).

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TABLE I. Examination and tests.

Inspection	Requirement	Test Method	First Article	Conformance
Visual and dimensional examination	3.2, 3.2.1, 3.3.1, 3.4.1, 3.5, 3.6, 3.7, 3.11, 3.12, 3.13, 3.14	4.4	X	X
Hydrostatic test	3.5, 3.8	4.5.1	X	X
Operational, room temperature tests	3.9	4.5.2, 4.5.2.1, 4.5.2.2	X	X
Operational, varying temperature test	3.9	4.5.3	X	--
Endurance test	3.4.2, 3.9	4.5.4	X	--
Performance check test	3.3.2, 3.4.2	4.5.5	X	X
Shock test	3.11	4.5.6	X	--

4.4 Visual and dimensional examination. Pump and motor units shall be visually and dimensionally examined to verify compliance with the requirements of this specification. If one or more defects are found in the sample, the first article must be resubmitted.

4.5 Tests.

4.5.1 Hydrostatic pressure test. Each sample unit shall be subjected to an internal hydrostatic pressure of 1,000 psi (69 bar) for 15 minutes. Leakage past any joints or seals or other evidence of failure shall be cause for rejection.

4.5.2 Operational tests. The operational tests in the following paragraphs shall be conducted at room temperature, 68 to 72 °F (20 to 22 °C).

4.5.2.1 Pumps. Each sample pump shall be operated at a discharge pressure of 900 psi (62 bar) for 10 minutes. This pressure shall be obtained by rotating the crank at a speed of 60 rpm.

4.5.2.2 Motors. Each sample motor shall operate with 900 psi (62 bar) hydraulic fluid at the inlet port and a drive shaft output torque of 60 foot-pounds (81 N•m) for 10 minutes.

4.5.3 Operational tests at varying temperatures and pressures. Performance characteristics shall be obtained with the temperature of the hydraulic fluid at 50, 100, 150, and 200 °F (10, 38, 66, and 93 °C), with the pump input varying up to 83 foot-pounds (112 N•m) and with the motor rotation both clockwise and counterclockwise. The discharge pressure of the pump shall not exceed 900 psi (62 bar).

4.5.4 Endurance tests. Pumps and motors shall be operated for 40 hours without leakage or other evidence of failure. The input to the pump shall be constant at 0.48 horsepower (358 watts) with the hydraulic fluid at room temperature. Motor rotation shall be reversed every 8 hours of operation. The unit shall then be performance check tested in accordance with 4.5.5.

4.5.5 Performance check test. Every pump and motor unit shall receive a check test of performance characteristics with the hydraulic fluid at 50 °F and at 200 °F (10 and 93 °C). The performance requirements of 3.3.2 and 3.4.2 shall be verified.

4.5.6 Shock test. The pump and motor unit, when mounted and under 900 psi (62 bar) hydraulic pressure, shall withstand mechanical shock when subjected to the type A shock test as specified in MIL-DTL-901 for grade A, class I equipment.

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4.5.6.1 Extension of shock testing. Where shock tests, as required by this specification, have previously been conducted and the equipment offered is the same as that previously tested, extension of shock testing as specified in MIL-DTL-901 shall be acceptable.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These pump and hydraulic motor units are intended for, but not limited to, use in hydraulic valve remote control systems to increase the fluid pressure delivered to the motors, which, when mounted on a valve, transforms the fluid pressure into rotary motion and transmits a torque to the valve stem. Such remote operators are used aboard Navy combatant and non-combatant vessels. Current stock of these pump and hydraulic motor units may be used until depleted; future requisitions for these pump and hydraulic motor units should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1).

c. Conformance tests if not as specified herein (see 4.3).

d. Packaging requirements (see 5.1).

e. Documentation, if required (see 6.4).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, samples selected from the first lot size of production items, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.5.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Suggested documentation. Technical manuals, drawings, and test reports should be requested as part of the acquisition of new pump and motor units under this specification (see 6.2).

6.5 Onboard repair parts. One set of gaskets, packing, and seals should be supplied as onboard repair parts for every two hydraulic units.

6.6 Subject term (key word) listing.

Crank

Hydrostatic

Positive displacement

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
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